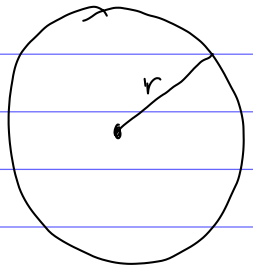
8



$A = A(t)$

$r = r(t)$

$\frac{dr}{dt} = 15.$ Find $\frac{dA}{dt}$ when $r = 50.$

$$A = \pi r^2$$

$$\frac{dA}{dt} = \pi \cdot 2r \cdot \frac{dr}{dt}$$

$$\frac{dA}{dt} = \pi \cdot 2 \cdot 50 \cdot 15$$

$$= \pi \cdot 100 \cdot 15$$

$$= 1500\pi$$

When the radius is 50, the area is increasing by 1500π $cm^2/sec.$